

A Complete Fourier Spectral Atlas for Constant- f Rotating Shallow Water

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Abstract

We give a global same-clock theorem for the linear rotating shallow-water equations on the doubly periodic 2π -torus. Each lattice mode is a three-by-three skew-Hermitian block. Its exact zero, plus, and minus projectors resolve the stationary geostrophic/potential-vorticity branch and the two inertia-gravity branches, including all zero-parameter faces. We also prove the sum-of-two-squares shell multiplicities, a necessary and sufficient periodicity test for finite Fourier support, and the global unitary but noncompact/non-Schatten boundary. A deterministic ledger and independent symbolic and numerical paths audit the claims. The model has constant f ; no beta-plane Rossby assertion is made.

1 Model and statement

Let $u = (u_1, u_2)$ and ϕ be complex fields on $\mathbb{T}^2$, and let

$$u_t + fJu + c\nabla\phi = 0, \quad \phi_t + c\nabla \cdot u = 0, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Here $f \in \mathbb{R}$, $c \geq 0$, and t is physical time. With $e_n(x) = e^{in \cdot x}$, $n \in \mathbb{Z}^2$, the coefficient vector $X_n = (\hat{u}_{1,n}, \hat{u}_{2,n}, \hat{\phi}_n)^T$ obeys

$$\dot{X}_n = G_n X_n, \quad G_n = \begin{pmatrix} 0 & f & -icn_1 \\ -f & 0 & -icn_2 \\ -icn_1 & -icn_2 & 0 \end{pmatrix}, \quad \omega_n = (f^2 + c^2|n|^2)^{1/2}.$$

For $c > 0$, the Fourier multiplier is taken on its maximal graph domain

$$D(G) = \left\{ X \in L^2 : \sum_{n \in \mathbb{Z}^2} \|G_n X_n\|_{\mathbb{C}^3}^2 < \infty \right\};$$

trigonometric polynomials form a core and H^1 is contained in this domain. The skew-Hermitian blocks close to a skew-adjoint generator and hence to the global unitary L^2 group. When $c = 0$, the bounded pointwise rotation is treated separately.

Theorem 1 (projector, shell, and operator atlas). *For every n with $\omega_n > 0$, $G_n^* = -G_n$,*

$$G_n^3 + \omega_n^2 G_n = 0, \quad P_0 = I + \frac{G_n^2}{\omega_n^2}, \quad P_{\pm} = \frac{1}{2} \left(-\frac{G_n^2}{\omega_n^2} \mp \frac{iG_n}{\omega_n} \right),$$

and P_0, P_+, P_- are mutually orthogonal projectors with ranks 1, 1, 1. Moreover

$$e^{tG_n} = P_0 + \cos(\omega_n t)(I - P_0) + \frac{\sin(\omega_n t)}{\omega_n} G_n.$$

The P_0 branch is stationary and is the linear potential-vorticity/geostrophic branch; $P_{\pm}$ are the inertia-gravity branches. At $n = 0$, $f \neq 0$ gives $0, \pm i|f|$, while $f = 0$ gives a zero block. For $c > 0$, the Fourier potential-vorticity quantity

$$\hat{\zeta}_n - \frac{f}{c}\hat{\phi}_n, \quad \hat{\zeta}_n = i(n_1\hat{u}_{2,n} - n_2\hat{u}_{1,n}),$$

is constant in time.

For $q > 0$, the shell $|n|^2 = q$ has $r_2(q) = 4(d_1(q) - d_3(q))$ modes. A finite Fourier-support solution is T -periodic if and only if $\omega_n T \in 2\pi\mathbb{Z}$ for every participating nonzero branch. The full group on $L^2(\mathbb{T}^2; \mathbb{C}^3)$ is unitary, but for every t , including $t = 0$, it is neither compact nor in any Schatten class.

Proof. Fourier transformation gives G_n . Direct multiplication yields the minimal polynomial $z(z^2 + \omega_n^2)$, and inspection of the entries gives $G_n^* = -G_n$. The three Lagrange polynomials for the roots $0, \pm i\omega_n$ are precisely P_0, P_+, P_- , proving orthogonality and the exponential after reducing its power series modulo the cubic. Taking $\zeta = \partial_1 u_2 - \partial_2 u_1$ gives $\zeta_t = -f\nabla \cdot u$, while $\phi_t = -c\nabla \cdot u$, proving the potential-vorticity identity for $c > 0$. The shell formula is the classical sum-of-two-squares identity. Applying the projectors to a finite superposition proves the periodicity equivalence. Finally, skew-adjointness gives unitarity; an orthonormal sequence of distinct Fourier modes is carried to another orthonormal sequence of norm one, proving noncompactness and the failure of every Schatten condition. $\square$

Source note

The f -plane doubly periodic normal-mode context is discussed by Mohebalhojeh and Dritschel [1]; Salmon [2] provides Hamiltonian projection context. Beta-plane results are intentionally outside the frozen model.

References

- [1] A. R. Mohebalhojeh and D. G. Dritschel, *Hierarchies of Balance Conditions for the f -Plane Shallow-Water Equations*, *Journal of the Atmospheric Sciences* 58 (2001), 2411–2426. DOI: 10.1175/1520-0469(2001)058<2411:HOBCT>2.0.CO;2.
- [2] R. Salmon, *Semigeostrophic theory as a Dirac-bracket projection*, *Journal of Fluid Mechanics* 196 (1988), 345–358. DOI: 10.1017/S0022112088002733.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic or operator claim. **Data and code.** The theorem, ledger, independent checks, and build instructions are released with HCS-C217. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked the displayed claims and metadata. This is not external peer review.